

Exact Repair Obstructions for a Sensitivity–14 Composition Built from the Kushilevitz Function

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Abstract

Eyal Kushilevitz’s six-variable Boolean function, recorded in a footnote of Nisan and Wigderson, has real degree three and full sensitivity six. We use a coordinate relabeling of that function as an outer map. Substituting four disjoint three-variable not-all-equal gates and two coordinate inputs gives an explicit Boolean function on fourteen variables with sensitivity fourteen and real degree exactly six. Its degree-six homogeneous component is $-E_1E_3(E_0 + E_2)$, consisting of exactly 54 monomials, each with coefficient -1 .

The main results exclude several precisely specified routes to lowering this construction to degree five. First, a degree-five, sensitivity-twelve base A , built from the same outer function, admits no Boolean function B on the same twelve variables satisfying $B(0) = 1$ and $\deg(B - A) \leq 4$. Second, a target-labelled repair cannot factor only through the four not-all-equal outputs and the two coordinate ports. Third, twisting any such factored function by the XOR parities of all four raw blocks always has degree greater than five. Full proofs rest on two direct-sum noncancellation lemmas: one for weighted top signatures under disjoint block substitution, and one for blockwise Walsh signatures.

We also give two exact, lossless parameterizations of all Boolean corrections relative to the fixed base P : an additive residual system and an XOR correction system. For the simplest XOR representative $J_0 = g_1g_3(g_0 \oplus g_2)$, one has $\deg_{\mathbb{R}} J_0 = 8$, while exact integer Möbius inversion shows that $P \oplus J_0$ has degree ten. Its top component is $4E_0E_1E_2E_3x_{12}x_{13}$, comprising exactly 81 monomials. A standalone verifier accompanies the paper. The results do not construct a degree-five, sensitivity-fourteen function and do not improve the Kushilevitz recursive exponent.

1 Introduction

For a Boolean function $f : \{0, 1\}^n \rightarrow \{0, 1\}$, let $\deg_{\mathbb{R}}(f)$ be the degree of its unique real multilinear polynomial and let $s(f)$ be its maximum sensitivity. Nisan and Szegedy established the foundational connection between real degree and combinatorial query measures [NS94]; Huang later proved the sensitivity conjecture [Hua19]. The explicit recursive benchmark used in this paper comes from a six-variable function due to Eyal Kushilevitz. Kushilevitz did not publish the function separately; it appears in footnote 1 of Nisan and Wigderson’s communication-complexity paper [NW95]. An explicit exposition is also given in [HKP11]. Its iterated composition has sensitivity 6^k and degree 3^k .

This paper begins from that exact historical object. Our first construction has the desired fourteen sensitive directions but degree six rather than five. Thus “near construction” has only the literal meaning “one degree above the target.” It does not itself improve any exponent. A degree-five, sensitivity-fourteen function would clear the Kushilevitz exponent because

$$3^{18} = 387420489 > 362797056 = 6^{11}, \quad 14^{11} = 4049565169664 > 3814697265625 = 5^{18}.$$

The function constructed here has parameters $(6, 14)$, not $(5, 14)$.

Our results are structural and confined to the fixed constructions below. We identify the complete degree-six layer of P , analyze selector and parity-twist obstructions, and parameterize every Boolean correction relative to P in two exact coordinate systems. These parameterizations are not ansatzes. The obstruction theorems exclude only the explicitly stated subfamilies within them.

Main results.

- (i) An explicit Kushilevitz-based Boolean function P on fourteen variables has sensitivity fourteen, degree six, and exactly 54 degree-six monomials, all with coefficient -1 (Theorem 5.1).
- (ii) A related degree-five, sensitivity-twelve base A has no arbitrary Boolean selector-neighbour B on the same variables with $B(0) = 1$ and $\deg(B - A) \leq 4$ (Theorem 6.1).
- (iii) No target-labelled degree-five repair can factor only through the four block outputs and two ports (Theorem 7.2).
- (iv) XOR twisting a gate-factored Boolean function by all four raw block parities always has degree greater than five (Theorem 7.4).
- (v) Additive and XOR correction systems are exact equivalences (Theorems 8.1 and 8.2). For the natural representative J_0 , the corrected function $P \oplus J_0$ has degree ten and an exact 81-term top layer (Theorem 9.1).

Provenance of proof and computation. The construction formulas, noncancellation lemmas, selector obstruction, gate-factor obstruction, parity-twist obstruction, and exact reformulations are proved symbolically. Exact finite computation is used only to materialize truth tables and coefficient tables and to verify the complete coefficient profile of $P \oplus J_0$. The top degree-ten formula also has a hand proof. The accompanying verifier uses only integer arithmetic.

2 Prior work and scope

The distinction between the historical base function and the results of this paper is important.

Kushilevitz and Nisan–Wigderson. The six-variable degree-three, sensitivity-six outer function is due to Eyal Kushilevitz and was recorded by Nisan and Wigderson [NW95]. Neither that function nor its recursive amplification is claimed here as new. Our displayed outer cubic is an explicit coordinate relabeling of it. Nisan and Wigderson identify their source as “E. Kushilevitz, private communication, 1994.” What is studied here is a different, nonrecursive substitution with four quadratic blocks and two linear ports, together with its repair equations and obstruction theory.

Amano. Amano proved that no fully sensitive Boolean function on ten variables has real degree at most four [Ama10, Theorem 3]. Restricting a function at a point with at least ten sensitive coordinates to ten such coordinates therefore gives the global consequence

$$\deg_{\mathbb{R}}(f) \leq 4 \implies s(f) \leq 9.$$

Neither this bound nor Amano’s ten-variable exclusion is claimed here as new. Amano also reported a computer-verified obstruction for the narrower class of fourteen-variable degree-five functions whose odd-degree coefficients lie in $\{0, 1\}$ and even-degree coefficients lie in $\{0, -1\}$ [Ama10, Fact 6]. That bounded-polynomial obstruction and our fixed-base obstructions are differently scoped; neither is asserted here to subsume the other. In later work, Amano enumerated, up to NPN equivalence, nondegenerate Boolean functions of sensitivity three and determined extremal values of several measures in that class [Ama17]. Our quadratic-rigidity proposition is only a local analytic statement under the additional real-degree-two hypothesis, not a new enumeration or a classification of all sensitivity-three functions.

Proskurin. Proskurin sharpened universal degree–block-sensitivity inequalities and derived the unique univariate polynomial that could arise by symmetrizing a hypothetical fully sensitive ten-variable degree-four function [Pro21]. Amano’s earlier theorem already excludes the existence of such a Boolean function. Our claims neither improve Proskurin’s universal inequalities nor provide a new global low-degree sensitivity bound; they concern specified twelve- and fourteen-variable Kushilevitz-based constructions.

Maitra–Mukherjee–Stănică–Tang. Maitra, Mukherjee, Stănică, and Tang connected real degree to resiliency, studied recursive amplification and higher-order sensitivity, and analyzed low-degree fully sensitive functions on small numbers of variables [MMST25]. Their computations include searches among rotation-symmetric functions through ten variables. Our fourteen-variable construction does not improve their or Kushilevitz’s separation exponent. The principal statements established in this manuscript include the explicit 54-term degree-six component, the two exact fixed-base parameterizations, and the selector, gate-factor, fixed-routing, parity-twist, and local alternating-sum obstructions with their stated hypotheses. A sufficiently broad priority search has not yet been completed, so we make no novelty or priority claim for these restricted statements.

3 Definitions and the Kushilevitz outer function

For $x \in \{0, 1\}^n$ and $i \in [n]$, write $x^{\oplus i}$ for x with coordinate i flipped. The local and maximum sensitivities are

$$s(f, x) = |\{i : f(x) \neq f(x^{\oplus i})\}|, \quad s(f) = \max_x s(f, x).$$

All unqualified degrees below are real multilinear degrees. The algebraic normal form $\text{ANF}(f)$ is the unique multilinear polynomial over $\mathbb{F}_2$; its degree will always be written $\deg_{\mathbb{F}_2}$, never simply “degree.” See [O’D14] for standard Boolean-function analysis conventions.

In zero-based coordinates, one standard form of the Kushilevitz cubic is

$$h_K(z) = \sum_{i=0}^5 z_i - \sum_{0 \leq i < j < 6} z_i z_j + \sum_{D \in \mathcal{D}_K} z_D,$$

where $z_D = \prod_{i \in D} z_i$ and

$$\mathcal{D}_K = \{023, 014, 034, 123, 124, 015, 025, 135, 245, 345\}.$$

This is the function attributed to Kushilevitz in [NW95, HKP11].

For the block placement used in this paper it is convenient to relabel its coordinates. Put

$$K(y_0, \dots, y_5) = h_K(y_0, y_1, y_2, y_5, y_3, y_4).$$

Equivalently,

$$K(y) = \sum_i y_i - \sum_{i < j} y_i y_j + \sum_{D \in \mathcal{D}} y_D,$$

with

$$\mathcal{D} = \{013, 014, 024, 025, 035, 123, 125, 145, 234, 345\}.$$

Equivalently, $(y_0, y_1, y_2, y_3, y_4, y_5) = (z_0, z_1, z_2, z_4, z_5, z_3)$.

Proposition 3.1 (Kushilevitz function). *The set $\mathcal{D}$ is a $2 - (6, 3, 2)$ design. The function K is Boolean, has degree exactly three, and satisfies $K(0) = 0$ and $K(e_i) = 1$ for all six coordinates.*

Proof. Direct inspection of the ten displayed triples shows that every pair occurs exactly twice. Hence every point occurs in five blocks. If $S \subseteq [6]$, $|S| = w$, and $b(S)$ is the number of design blocks contained in S , then

$$K(1_S) = w - \binom{w}{2} + b(S).$$

For $w = 0, 1, 2$ the values are $0, 1, 1$. For $w = 3$, the value is the indicator that S is a block. If $w = 4$, the complementary pair is met by $5 + 5 - 2 = 8$ blocks, leaving two blocks inside S ; the value is $4 - 6 + 2 = 0$. If $w = 5$, exactly five blocks avoid the omitted point, so the value is $5 - 10 + 5 = 0$. Finally, at $w = 6$ it is $6 - 15 + 10 = 1$. Thus K is Boolean. Its ten cubic coefficients are nonzero, so its degree is three, and the statements at the origin and singleton points are immediate. $\square$

4 Two block-signature noncancellation lemmas

The obstruction proofs use two different direct-sum facts. We state both in full because conflating them conceals essential hypotheses. We adopt the conventions $\deg 0 = -\infty$ and $\max \emptyset = -\infty$.

Let $X_i = \{x_{i0}, x_{i1}, x_{i2}\}$ be pairwise disjoint triples and define

$$L_i = x_{i0} + x_{i1} + x_{i2}, \quad E_i = x_{i0}x_{i1} + x_{i0}x_{i2} + x_{i1}x_{i2}, \quad g_i = L_i - E_i = \text{NAE}_3(X_i).$$

Lemma 4.1 (Weighted top signatures). *Let $z_1, \dots, z_s$ be distinct raw coordinates disjoint from all blocks, and let*

$$Q(y, z) = \sum_{A \subseteq [r]} \sum_{B \subseteq [s]} c_{A,B} y_A z_B$$

be the unique multilinear meta polynomial over a characteristic-zero field. Give each y_i weight two and each z_j weight one. For every A, B , and every choice of a pair $p_i \in \binom{X_i}{2}$ for $i \in A$, the coefficient in $Q(g, z)$ of

$$x_{\cup_{i \in A} p_i} z_B$$

is exactly $(-1)^{|A|} c_{A,B}$. Consequently,

$$\deg Q(g, z) = \max_{c_{A,B} \neq 0} (2|A| + |B|).$$

Proof. The displayed raw monomial contains exactly two variables from each block in A , none from any block outside A , and exactly the ports in B . From the meta monomial $y_A z_B$, it is obtained only by choosing one quadratic term from each $-E_i$, and hence has coefficient $(-1)^{|A|} c_{A,B}$.

No different meta support can contribute to the same monomial. A block in $A' \setminus A$ would contribute at least one raw variable; a block in $A \setminus A'$ would contribute none; and a different port set has a different raw port support. Thus these top signatures lie in pairwise disjoint monomial

subspaces and cannot cancel. A nonzero coefficient of maximum weighted degree therefore produces a nonzero raw monomial of that degree, while no substitution can exceed that weighted degree. This proves the formula. $\square$

Remark 4.2. Disjointness and prior multilinearization are necessary. If two formal gates use the same raw block, $y_0 - y_1$ may vanish after substitution. If a meta polynomial is not multilinearized, $y_0^2 - y_0$ is an immediate counterexample. The lemma also says nothing about a function that sees raw orientation inside a block rather than only g_i .

For the second lemma use spin variables $s_{ij} = 1 - 2x_{ij}$, and put

$$\chi_i = s_{i0}s_{i1}s_{i2}, \quad u_i = \frac{s_{i0}s_{i1} + s_{i0}s_{i2} + s_{i1}s_{i2}}{2}, \quad r_i = 1 - 2g_i = u_i - \frac{1}{2}.$$

Lemma 4.3 (Walsh block signatures). *Let $t_1, \dots, t_s$ be port spins disjoint from the four blocks, and write a multilinear meta-spin polynomial, after the shift $r_i = u_i - \frac{1}{2}$, as*

$$q(r, t) = \sum_{U \subseteq [4]} \sum_{B \subseteq [s]} d_{U,B} u_U t_B.$$

After multiplication by $\chi = \chi_0\chi_1\chi_2\chi_3$, every nonzero $d_{U,B}$ produces $3^{|U|}$ distinct Walsh characters of degree

$$12 - 2|U| + |B|,$$

each with coefficient $2^{-|U|}d_{U,B}$. Characters produced by distinct (U, B) , or by different coordinate choices inside a block, are distinct. Hence

$$\deg(\chi q) = \max_{d_{U,B} \neq 0} (12 - 2|U| + |B|).$$

Proof. For $i \in U$,

$$\chi_i u_i = \frac{s_{i0} + s_{i1} + s_{i2}}{2},$$

so block i contributes one of three degree-one characters. For $i \notin U$, it contributes the unique degree-three character χ_i . The ports contribute exactly the character indexed by B .

The blockwise signature “one selected spin” versus “all three spins” recovers U , the selected spin recovers the coordinate choice, and the port support recovers B . The corresponding Walsh characters are therefore pairwise distinct. Since Walsh characters form a basis, no cross-signature cancellation is possible. Summing the four block degrees and the port degree gives the stated degree formula. $\square$

5 The degree-six, sensitivity-fourteen composition

Use four disjoint blocks $X_i = (x_{3i}, x_{3i+1}, x_{3i+2})$, $0 \leq i < 4$, and define

$$P = K(g_0, g_1, g_2, g_3, x_{12}, x_{13}).$$

Theorem 5.1. *The function P is Boolean, satisfies*

$$P(0) = 0, \quad P(e_j) = 1 \quad (0 \leq j < 14),$$

and therefore has sensitivity fourteen at the origin. Its degree is exactly six, and its homogeneous degree-six part is

$$P_6 = -E_0E_1E_3 - E_1E_2E_3 = -E_1E_3(E_0 + E_2).$$

It contains exactly 54 degree-six monomials, each with coefficient -1 , and has no term of higher degree.

Proof. Booleanity follows by composing Boolean functions. At the raw origin all six outer inputs vanish. A singleton in one of the first twelve coordinates changes exactly one g_i from zero to one; the final two coordinates are the two remaining outer singleton directions. The target rows follow from Proposition 3.1.

By Lemma 4.1, raw degree six can arise only from an outer cubic containing three heavy inputs. In $\mathcal{D}$ the only such triples are 013 and 123, whose top terms are respectively $-E_0E_1E_3$ and $-E_1E_2E_3$. Their block signatures differ, so they cannot cancel. Each product of three E_i 's has $3^3 = 27$ distinct monomials; the two sets are disjoint. No outer term has weighted degree above six. $\square$

6 The central selector obstruction

The following twelve-variable slice is a natural attempt to obtain a degree-five base and add one or two selector variables. Use three NAE blocks on $x_0, \dots, x_8$, put $\ell_3 = x_9, \ell_4 = x_{10}, \ell_5 = x_{11}$, and define

$$A = K(g_0, g_1, g_2, \ell_3, \ell_4, \ell_5).$$

Theorem 6.1 (Arbitrary-selector obstruction). *The function A is Boolean, has sensitivity twelve at the origin, and has degree exactly five. Its degree-five part is*

$$A_5 = E_0E_1(\ell_3 + \ell_4) + E_0E_2(\ell_4 + \ell_5) + E_1E_2(\ell_5 + \ell_3),$$

which consists of 54 distinct coefficient-+1 monomials.

There is no Boolean function $B : \{0, 1\}^{12} \rightarrow \{0, 1\}$ satisfying

$$B(0) = 1, \quad \deg(B - A) \leq 4.$$

Proof. The heavy triple 012 is not a block of $\mathcal{D}$, so no outer cubic contains three heavy inputs. The six heavy-heavy-linear blocks are

$$013, 014, 024, 025, 125, 123.$$

The weighted-signature lemma yields the displayed degree-five part. Each of its six summands contributes nine distinct masks, distinguished by their heavy block pair and linear port. The Booleanity and twelve singleton rows follow as in Theorem 5.1.

Assume that B exists. Average B under independent coordinate permutations inside each heavy block and under independently complementing all three coordinates of each block. Call the average C . These affine cube automorphisms preserve real degree and fix A ; therefore

$$0 \leq C \leq 1, \quad \deg(C - A) \leq 4.$$

The action on one raw block has exactly two orbits, the points with NAE value zero and those with NAE value one. Thus C factors through the three NAE outputs and three linears. The meta-origin

orbit contains the eight points obtained by choosing each raw block to be 000 or 111. Since it contains the raw origin and $B(0) = 1$,

$$c_0 = C(0, 0, 0, 0, 0, 0) \geq \frac{1}{8}.$$

Apply Lemma 4.1 to the unique multilinear meta polynomial of $C - A$. Every meta coefficient of weight greater than four vanishes. Consequently all HHH coefficients of C , including their linear supersets, are zero; the six HHL coefficients agree with those of A and equal one; and all HHLL and HHLLL coefficients vanish.

Fix a linear assignment $z \in \{0, 1\}^3$, and write h_z for C as a function of the three heavy variables. It is quadratic. If $b_{ij}(z)$ are its three heavy-pair coefficients, the preceding coefficient identities give

$$\begin{aligned} b_{01}(z) &= \beta_{01} + z_3 + z_4, \\ b_{02}(z) &= \beta_{02} + z_4 + z_5, \\ b_{12}(z) &= \beta_{12} + z_5 + z_3. \end{aligned}$$

For a quadratic polynomial on three variables define

$$S_z = h_z(111) - h_z(100) - h_z(010) - h_z(001) + 2h_z(000).$$

Möbius inversion on the three-cube shows that S_z is the sum of the three pair coefficients. Hence

$$S_{111} = S_{000} + 6.$$

Because every value of h_z lies in $[0, 1]$,

$$S_{000} \geq -3 + 2c_0, \quad S_{111} \leq 3.$$

But $c_0 \geq 1/8$ implies

$$S_{111} \geq -3 + \frac{1}{4} + 6 = \frac{13}{4} > 3,$$

a contradiction. □

Remark 6.2 (Exact scope). The theorem treats arbitrary Boolean B on the same twelve raw variables; it is not a restriction to gate-factored selectors. It depends essentially on the fixed base A , the anchored value $B(0) = 1$, and the degree-four difference. For example, without $B(0) = 1$ one may take $B = A$; without Booleanity one may take $B = A + 1$. It does not exclude a different base, additional variables, or a different outer architecture.

7 Gate-factor and parity-twist obstructions

We next return to the fourteen-variable setting. These results explain why a successful repair cannot see only the four NAE output bits and why the most obvious full-block parity dualization also fails.

Proposition 7.1 (Quadratic rigidity). *Every Boolean function of real degree at most two has sensitivity at most three. More precisely, if $h(0) = 0$ and its sensitive set at the origin is exactly $\{a, b, c\}$, then*

$$h = \text{NAE}_3(x_a, x_b, x_c)$$

and h is independent of every other coordinate.

Proof. Translate a maximally sensitive point to the origin and complement the output if necessary; these operations preserve degree. Write

$$h = \sum_i \alpha_i x_i + \sum_{i < j} \beta_{ij} x_i x_j.$$

Every sensitive coordinate has $\alpha_i = 1$. On a pair of sensitive coordinates, Booleanity gives $2 + \beta_{ij} \in \{0, 1\}$, so $\beta_{ij} \in \{-2, -1\}$. On any sensitive triple, a coefficient -2 would force the triple value to be negative; hence all three pair coefficients are -1 . Four sensitive coordinates are then impossible, because their all-one face value would be $4 - \binom{4}{2} = -2$.

In the equality case, the restriction to a, b, c is NAE. For an outside coordinate k , each β_{ik} with $i \in \{a, b, c\}$ belongs to $\{-1, 0\}$. Evaluating at $e_a + e_b + e_c + e_k$ forces all three to zero. For two outside coordinates k, ℓ , Booleanity gives $\beta_{k\ell} \in \{0, 1\}$, while evaluation at $e_a + e_k + e_\ell$ forces it to zero. $\square$

Theorem 7.2 (Gate-factor obstruction). *Let $T : \{0, 1\}^6 \rightarrow \{0, 1\}$ be arbitrary and put*

$$F = T(g_0, g_1, g_2, g_3, x_{12}, x_{13}),$$

where the four raw blocks and the two coordinate ports are pairwise disjoint. If $F(0) = 0$ and $F(e_j) = 1$ for every $0 \leq j < 14$, then $\deg F > 5$.

Proof. All 64 meta inputs are independently attainable, so T has a unique multilinear meta representation. If $\deg F \leq 5$, Lemma 4.1 forces every meta coefficient of weight above five to vanish. Set both ports to zero. The remaining Boolean function of g_0, g_1, g_2, g_3 has meta degree at most two, value zero at its origin, and value one at all four meta singletons. This contradicts Proposition 7.1. $\square$

Remark 7.3. The target labels are essential: the constant function is a gate-factored counterexample without them. The theorem does not cover raw orientation dependence inside a block.

Let $p_i = x_{i0} \oplus x_{i1} \oplus x_{i2}$ denote XOR parity on raw block i .

Theorem 7.4 (Parity twist). *For every Boolean meta function $T(g_0, g_1, g_2, g_3, x_{12}, x_{13})$, the Boolean function*

$$F = T \oplus p_0 \oplus p_1 \oplus p_2 \oplus p_3$$

has real degree greater than five.

Proof. Passing between bit variables and spin variables preserves the degree filtration. Let $q_T = 1 - 2T$. Expand q_T uniquely in the shifted block variables $r_i = u_i - \frac{1}{2}$ and the two port spins. The output spin of F is

$$q_F = q_T \chi_0 \chi_1 \chi_2 \chi_3.$$

If $\deg q_F \leq 5$, Lemma 4.3 forces every sector with $|U| \leq 3$ to vanish, since its core degree is at least six. In the only remaining sector $U = [4]$, the coefficient has port degree at most one. Thus

$$q_T = a(t_{12}, t_{13}) u_0 u_1 u_2 u_3$$

for an affine function a of the port spins.

Fix the ports. If $a = 0$, then $q_T = 0$, already contradicting Booleanity. Otherwise the right side has magnitude $|a| \prod_i |u_i|$. For one block, $|u_i| = 3/2$ on an all-equal raw orientation and $|u_i| = 1/2$ on a non-all-equal orientation. All sixteen meta patterns are attainable, so the magnitude varies with the block outputs. This contradicts Booleanity, which requires $|q_T| = 1$ everywhere. $\square$

Remark 7.5 (Exact scope). This theorem concerns a meta function followed by the parity twist on all four blocks. It proves nothing about a partial twist, a nonlinear orientation correction, or a T that already sees raw orientations. Indeed, if raw orientation is allowed, choosing $T = p_0 \oplus p_1 \oplus p_2 \oplus p_3$ gives the constant zero function after twisting.

8 Two exact reformulations of every repair

The preceding theorems exclude particular structural classes. We now record two descriptions that exclude nothing. They are changes of coordinates on the full set of Boolean functions relative to the fixed function P , not search restrictions.

Let

$$\mathcal{T} = \{0, e_0, e_1, \dots, e_{13}\}$$

be the fifteen target rows, and put

$$\Delta = E_1 E_3 (E_0 + E_2) = -P_6.$$

Theorem 8.1 (Exact additive residual system). *Let $R : \{0, 1\}^{14} \rightarrow \mathbb{R}$, let the same symbol denote its unique real multilinear polynomial, and set $F = P + R$. Then F is Boolean, has degree at most five, and agrees with P on $\mathcal{T}$ if and only if all three of the following hold:*

$$R = \Delta + L \quad \text{for a multilinear polynomial } \deg L \leq 5, \quad (1)$$

$$R(x)(R(x) + 2P(x) - 1) = 0 \quad \text{for every } x \in \{0, 1\}^{14}, \quad (2)$$

$$R(t) = 0 \quad \text{for every } t \in \mathcal{T}. \quad (3)$$

Consequently, every admissible Boolean F occurs exactly once, with $R = F - P$.

Proof. Since P has no terms above degree six and $P_6 = -\Delta$, cancellation of every term of degree at least six in $P + R$ is equivalent to Eq. (1). This proves the degree equivalence, including the absence of any term of degree greater than six in R .

On a cube point, $P^2 = P$. Therefore

$$F(F - 1) = (P + R)(P + R - 1) = R(R + 2P - 1).$$

The left side vanishes exactly when $F \in \{0, 1\}$, proving Eq. (2). More explicitly, where $P = 0$, the permitted residual values are $R \in \{0, 1\}$, and where $P = 1$, they are $R \in \{0, -1\}$. Finally, $F(t) = P(t)$ is exactly $R(t) = 0$, which proves Eq. (3). The identity $R = F - P$ proves uniqueness. $\square$

All three conditions are indispensable. The residuals $R = -P$ and $R = 1 - P$ satisfy the Boolean equation but give the constant-zero and constant-one functions, so they demonstrate the need for the target-row condition. Conversely, $R = \Delta$ satisfies the high-degree cancellation and vanishes on $\mathcal{T}$, but at $x = e_0 + e_1 + e_3 + e_4 + e_9 + e_{10}$ one has $P(x) = \Delta(x) = 1$, hence $(P + R)(x) = 2$. This demonstrates the need for the Boolean equation.

Theorem 8.2 (Exact XOR correction system). *For every Boolean $F : \{0, 1\}^{14} \rightarrow \{0, 1\}$ there is a unique Boolean mask*

$$J = P \oplus F,$$

and pointwise, as an identity over the reals,

$$F = P + (1 - 2P)J. \quad (4)$$

Thus the Boolean functions of degree at most five that agree with P on $\mathcal{T}$ are in bijection with the Boolean masks J satisfying

$$J|_{\mathcal{T}} = 0, \quad \deg_{\mathbb{R}}(P + (1 - 2P)J) \leq 5. \quad (5)$$

Every such mask necessarily satisfies

$$\text{ANF}_{\geq 6}(J) = \text{ANF}_{\geq 6}(P), \quad (6)$$

which here consists of exactly the 54 degree-six masks in $E_1 E_3 (E_0 + E_2)$, and no higher mask.

Proof. The first assertion is the pointwise identity $a \oplus b = a + (1 - 2a)b$ for $a, b \in \{0, 1\}$. It also gives $J = P \oplus F$, so the representation is unique. Equality of the target rows is equivalent to $J = 0$ there, proving the exact bijection Eq. (5).

For a Boolean function, reducing every integer coefficient of its real Möbius expansion modulo two gives its $\mathbb{F}_2$ -algebraic normal form. If $\deg_{\mathbb{R}} F \leq 5$, all real coefficients of F on masks of size at least six vanish and therefore so do the corresponding ANF coefficients. Since $F = P \oplus J$, its ANF is the sum of the ANFs of P and J . This proves Eq. (6). The description of the high masks follows from Theorem 5.1 after reduction of -1 modulo two. $\square$

Equation (6) immediately shows that any successful J depends on raw variables from all four NAE blocks. It is nevertheless only a necessary parity condition; real-degree cancellation is stricter.

Define the compact Boolean mask

$$J_0 = g_1 g_3 (g_0 \oplus g_2). \quad (7)$$

Its degree-six ANF layer is precisely $E_1 E_3 (E_0 + E_2)$, and it has no higher ANF term. Hence every mask satisfying Eq. (6) can be written uniquely as

$$J = J_0 \oplus W, \quad \deg_{\mathbb{F}_2} W \leq 5.$$

This is again a reparameterization, not an assertion that W is local, factored, or otherwise simple. Theorems 7.2 and 7.4 explain why two especially natural simplifications do not suffice.

9 The exact failure of the simplest mask J_0

The degree of J_0 must not be confused with the degree of the repaired function. The former is eight; the latter is ten.

Theorem 9.1 (The function $P \oplus J_0$ has degree ten). *The real multilinear degree of J_0 is eight, with top component*

$$(J_0)_8 = -2E_0 E_1 E_2 E_3.$$

For

$$F_0 = P \oplus J_0$$

the real multilinear degree is ten, and

$$(F_0)_{10} = 4E_0 E_1 E_2 E_3 x_{12} x_{13}.$$

Thus F_0 has exactly $3^4 = 81$ degree-ten monomials, each with coefficient $+4$.

Proof. At the six meta inputs $y = (y_0, y_1, y_2, y_3, y_4, y_5)$, the real polynomial of J_0 is

$$j = y_1 y_3 (y_0 + y_2 - 2y_0 y_2).$$

The unique weight-eight term after substituting the four heavy NAE gates is $-2g_0 g_1 g_2 g_3$, whose top component is $-2E_0 E_1 E_2 E_3$. Weighted top-signature noncancellation proves that it is nonzero and that no term has larger degree.

Multilinearizing the meta function $K \oplus j = K + j - 2Kj$ gives

$$\begin{aligned} & K - y_0 y_1 y_3 - y_1 y_2 y_3 + 2y_0 y_1 y_2 y_3 \\ & + 2y_0 y_1 y_3 y_4 + 2y_1 y_2 y_3 y_4 + 2y_0 y_1 y_3 y_5 + 2y_1 y_2 y_3 y_5 \\ & - 4y_0 y_1 y_2 y_3 y_4 - 4y_0 y_1 y_2 y_3 y_5 \\ & - 2y_0 y_1 y_3 y_4 y_5 - 2y_1 y_2 y_3 y_4 y_5 + 4y_0 y_1 y_2 y_3 y_4 y_5. \end{aligned}$$

Only the final meta monomial has weighted degree ten. Substituting $y_i = g_i$ for $0 \leq i < 4$, $y_4 = x_{12}$, and $y_5 = x_{13}$, its top signature is $4E_0 E_1 E_2 E_3 x_{12} x_{13}$. The weighted-signature lemma proves both noncancellation and the absence of any higher term. Each E_i offers three pair choices on a disjoint block, giving exactly $3^4 = 81$ monomials. $\square$

For completeness, exact integer Möbius inversion gives the following full nonzero coefficient profile. An entry $c : m$ means that exactly m monomials of that degree have coefficient c .

degree	number nonzero	coefficient multiplicities
1	14	+1 : 14
2	91	-1 : 91
3	192	+1 : 192
4	357	-1 : 168, +2 : 189
5	918	-4 : 162, -2 : 702, +1 : 54
6	1701	+2 : 972, +4 : 729
7	1890	-2 : 594, -4 : 1296
8	1269	+2 : 135, +4 : 1134
9	486	-4 : 486
10	81	+4 : 81

The total is 6999 nonzero coefficients. The table is a reproducible finite calculation, whereas the degree-ten and 81-term assertions in Theorem 9.1 are proved independently of it.

10 Further exact obstructions and residual constraints

The central no-go theorems already show that a successful correction, if one exists, cannot be a selector extension of A , a target-labelled function of the six meta outputs alone, or a full four-block parity twist of such a meta function. Two additional consequences clarify the remaining scope.

Corollary 10.1 (Rigidity of the fixed outer routing). *For $0 \leq i < 4$, let h_i be a Boolean function of real degree at most two that vanishes at its own origin and is sensitive there in all three coordinates of a designated raw-coordinate triple X_i . Assume the X_i are pairwise disjoint and route h_i to input i of K . Route two further raw coordinates, distinct from one another and from every X_i , to inputs 4 and 5. Then, after relabeling within blocks, the four quadratic gates are exactly the four NAE gates used in P . The composite therefore has the nonzero degree-six layer in Theorem 5.1 and is sensitive in all fourteen designated directions.*

Proof. Each quadratic gate must be sensitive in all three of its designated coordinates. Proposition 7.1 makes it NAE on exactly that triple and independent of every other coordinate. The claimed composite is therefore P up to within-block relabeling, which preserves its degree-six top signatures. $\square$

There is also a local constraint on every exact correction. For either outer design block 013 or 123, choose one of the three quadratic monomials from each associated E_i . Fix all other raw variables, including both ports, to zero. On the resulting six-dimensional face, each restricted NAE gate is OR_2 , and the restriction of P is OR_6 . There are 27 faces of each block signature, hence 54 such faces.

Proposition 10.2 (Alternating zero-set constraint). *Let $F : \{0, 1\}^6 \rightarrow \{0, 1\}$ have degree at most five, with $F(0) = 0$, and let*

$$Z = \{x \neq 0 : F(x) = 0\}.$$

Then

$$\sum_{x \in Z} (-1)^{|x|} = -1.$$

Consequently, every degree-five target-preserving correction of P must satisfy this identity on each of the 54 OR_6 faces above.

Proof. The coefficient of the full six-variable monomial is

$$\sum_{x \in \{0, 1\}^6} (-1)^{6-|x|} F(x) = \sum_{x \in \{0, 1\}^6} (-1)^{|x|} F(x).$$

It vanishes because $\deg F \leq 5$. Also $\sum_{x \in \{0, 1\}^6} (-1)^{|x|} = 0$, while the origin is not in Z . Subtracting the zero-set contribution from the full alternating sum proves the formula. Restriction cannot increase degree, so it applies to every listed face. $\square$

This condition is necessary, not sufficient. In particular, it does not replace either exact system in Section 8.

11 Standalone exact verifier

The supplementary script `verify_exact.py` is standalone, uses only the Python standard library, and has no numerical or symbolic-algebra dependency. It constructs K, P, A, J_0 , and $P \oplus J_0$ from the displayed formulas. For a truth vector $v[0], \dots, v[2^n - 1]$ in numeric mask order, it performs the in-place integer subset transform

$$c \leftarrow v; \quad c[m] \leftarrow c[m] - c[m \setminus \{i\}] \quad (i = 0, \dots, n - 1; i \in m).$$

The resulting $c[m]$ are exactly the real multilinear Möbius coefficients; no division or floating-point operation occurs.

The verifier computes every real multilinear coefficient and checks the claimed Booleanity, origin sensitivities, degrees, top supports, and the complete coefficient histogram of $P \oplus J_0$. Its deterministic summary is:

function	variables	degree	origin sensitivity	nonzero coefficients	top layer
K	6	3	6	31	ten coefficient-+1 cubics
A	12	5	12	421	54 coefficients +1
P	14	6	14	951	54 coefficients -1
J_0	14	8	0	1728	81 coefficients -2
$P \oplus J_0$	14	10	14	6999	81 coefficients +4

The script exits nonzero upon failure of any encoded assertion. The diagnostic row for J_0 is included specifically to prevent the erroneous statement $\deg J_0 = 10$: the degree-ten object is $P \oplus J_0$.

12 What is and is not resolved

The two parameterizations are exact and complete relative to the fixed base P , but the obstruction theorems exclude only their stated subfamilies. They leave open whether there exists a Boolean mask J , equivalently an additive residual R , satisfying the systems in Theorems 8.1 and 8.2. Any such correction must obey the 54 forced ANF masks, the target rows, the real Boolean equations, the degree cancellation, and the local alternating constraints. It must also evade the selector, gate-factor, fixed-routing, and full-parity-twist theorems. These facts point to raw orientation-dependent interactions across blocks, but they do not constitute an existence theorem, a completeness result for any smaller family, or a lower bound for all degree-five functions.

In particular, this paper proves no degree-five, sensitivity-fourteen construction and therefore no new asymptotic separation. The phrase “near-solution” refers only to the explicit degree-six function P and its explicitly computed degree-six homogeneous component.

Generative-AI disclosure

Generative-AI systems were used extensively for conjecture generation, symbolic exploration, code generation, proof drafting, internal auditing, and manuscript editing. The named human authors reviewed the final theorem statements, proofs, computations, and references and accept responsibility for the contents.

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